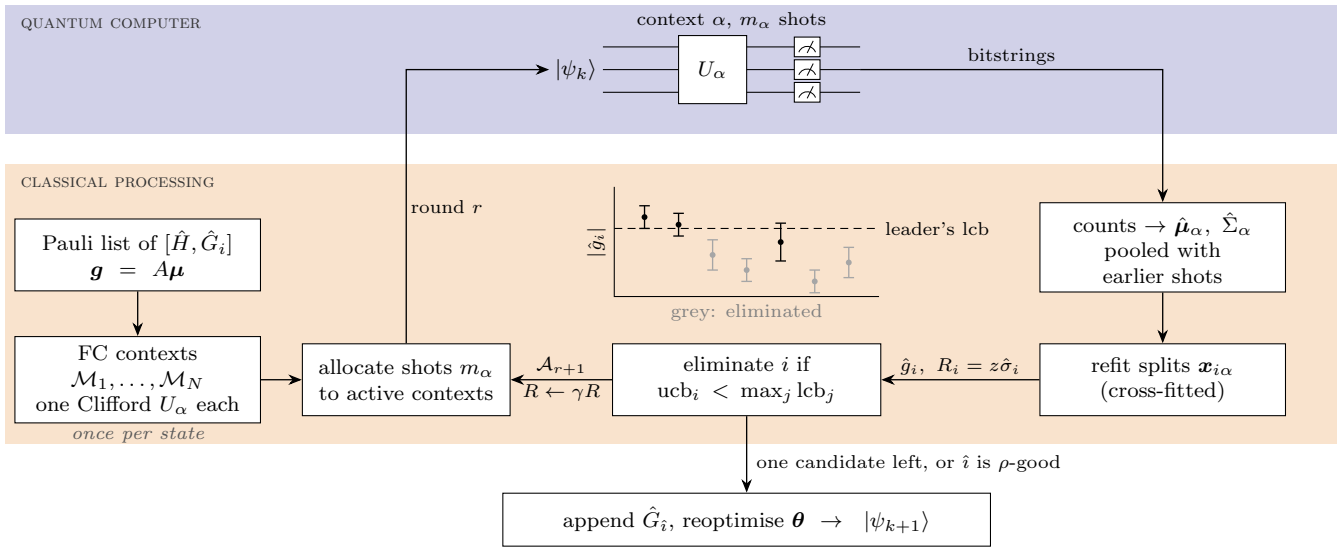

QUANTUM COMPUTER
context α, m_α shots
$|\psi_k\rangle$
U_α
bitstrings
CLASSICAL PROCESSING
round r
Pauli list of $[\hat{H}, \hat{G}_i]$
$\boldsymbol{g} = A\boldsymbol{\mu}$
$|\hat{g}_i|$
leader's lcb
grey: eliminated
counts $\rightarrow \hat{\boldsymbol{\mu}}_\alpha$, $\hat{\Sigma}_\alpha$
pooled with
earlier shots
FC contexts
$\mathcal{M}_1, \ldots, \mathcal{M}_N$
one Clifford U_α each
once per state
allocate shots m_α
to active contexts
$\mathcal{A}_{r+1}$
$R \leftarrow \gamma R$
eliminate i if
$\mathrm{ucb}_i < \max_j \mathrm{lcb}_j$
$\hat{g}_i,\ R_i = z\hat{\sigma}_i$
refit splits $\boldsymbol{x}_{i\alpha}$
(cross-fitted)
one candidate left, or $\hat{\imath}$ is ρ-good
append $\hat{G}_{\hat{\imath}}$, reoptimise $\boldsymbol{\theta}$ $\rightarrow$ $|\psi_{k+1}\rangle$